

SAFETY PLAN GENERATOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Of course. As STRUCTMIND AI, I will generate a comprehensive Lifting, Rigging, and Erection Safety Plan based on the project parameters provided, in accordance with OSHA 1926 Subpart R.

Steel Erection Safety Plan: 3-Storey Structure

Project Name: [Insert Project Name] Location: [Insert Project Address] Date Prepared: [Insert Date]
Prepared By: STRUCTMIND AI (To be reviewed and signed by a Qualified Person)

Disclaimer: This safety plan is a template based on the information provided. A project-specific plan must be developed, reviewed, and implemented by a Qualified Person on-site. All activities must comply with manufacturer specifications, engineering drawings, and all applicable federal, state, and local regulations, including OSHA 1926 Subpart R.

1. Lifting Plan Notes

This section outlines the requirements for all crane and hoisting operations.

1.1. Crane & Load Details:

- Crane Type: A mobile hydraulic crane with a minimum capacity of 60-80 tons is anticipated to meet the lift requirements. The final selection must be based on the specific crane's load chart.
- Heaviest Pick: Spandrel Girder
- Weight: 4.2 tons (8,400 lbs). Add 10-15% for rigging, headache ball, etc., for a total calculated load of ~9,500 lbs.
- Maximum Lift Radius: 70 feet.
- Load Chart Verification: The crane operator and lift director MUST verify that the calculated load at the 70-foot radius is well within the crane's rated capacity as per the manufacturer's load chart for its specific configuration (outrigger spread, counterweights, boom angle). No lift shall exceed 85% of the crane's rated capacity for that configuration.

1.2. Pre-Lift Requirements & Operations:

- Crane Setup:

- The crane will be set up on stable, compacted ground, confirmed by a competent person.
- Full outrigger extension and proper cribbing/pads are mandatory for all lifts.
- A swing path analysis will be conducted daily to ensure no overhead obstructions (especially power lines) or ground personnel are in the lift zone. A minimum clearance of 20 feet must be maintained from energized power lines.
- Inspections:
 - The crane operator will perform a documented daily inspection of the crane.
 - All rigging will be visually inspected by a Qualified Rigger prior to each shift. Damaged rigging will be immediately removed from service.
 - Weather: Lifting operations will cease when wind speeds exceed 25 mph or as specified by the crane manufacturer, whichever is lower. Operations will also cease during electrical storms or other adverse weather conditions.
 - Communication: Clear and constant communication via hand signals (per OSHA 1926.1400) or dedicated radio channels will be maintained between the operator, signal person, and riggers. The signal person must be qualified.
 - Pre-Lift Meeting: A pre-lift meeting will be held with the entire erection crew (operator, signal person, riggers, connectors) before the start of each day and for any non-routine lift to review the plan, hazards, and individual responsibilities.

2. Rigging Plan

This section details the rigging to be used, focusing on the heaviest pick.

2.1. Rigging for 4.2-ton Spandrel Girder:

- Configuration: A two-point vertical pick using a spreader bar is required to control the load and prevent crushing the girder flanges.
- Spreader Bar: A properly rated and certified spreader bar of sufficient length to ensure slings remain near-vertical (80-90 degrees) shall be used.
- Slings (Top): Two (2) wire rope or chain slings connecting the crane hook to the spreader bar. Each sling must have a vertical Working Load Limit (WLL) of at least 5 tons.
- Slings (Bottom): Two (2) wire rope choker slings or synthetic web slings (with edge protection) connecting the spreader bar to the girder. Each sling must have a vertical WLL of at least 5 tons. Slings will be basket-hitched around the girder at engineered pick points.
- Shackles: All shackles must be alloy steel, properly sized for the slings, with a WLL greater than the anticipated load. Screw-pin shackles must have the pin fully seated and secured.

2.2. General Rigging Practices:

- **Qualified Rigger:** All rigging must be selected and attached by a qualified rigger.
- **Taglines:** At least two (2) non-conductive taglines will be used on all column and girder lifts to control load movement.
- **Sling Protection:** Edge protection ("softeners") must be used whenever slings are in contact with sharp edges of steel.
- **Center of Gravity:** Rigging shall be attached above the load's center of gravity to ensure stability.

3. Fall Protection Plan (OSHA 1926 Subpart R)

Fall protection is mandatory for all personnel working at heights.

- **General Requirement:** All employees on a walking/working surface with an unprotected side or edge more than 15 feet above a lower level shall be protected by guardrail systems, safety net systems, or a Personal Fall Arrest System (PFAS).
- **Connectors (15 to 30 feet):** While working as a connector, employees between 15 and 30 feet must be provided with and wear a complete PFAS. They must be tied off when not actively receiving or connecting members.
- **All Personnel (Above 30 feet):** All personnel, including connectors, must be tied off using a PFAS 100% of the time.
- **Personal Fall Arrest System (PFAS):**
 - **A - Anchorage:** Anchor points must be independent of any platform support and capable of supporting 5,000 lbs. per person attached. Beam clamps, beam straps, or other engineered anchor points will be utilized.
 - **B - Body Harness:** Full-body harnesses will be worn by all personnel working at height.
 - **C - Connecting Device:** Lanyards (shock-absorbing) or Self-Retracting Lifelines (SRLs) will be used. Lanyards will be selected to ensure fall distance does not exceed 6 feet.
- **Controlled Decking Zone (CDZ):**
 - A CDZ may be established during metal decking installation over 15 feet.
 - Access will be limited to trained and authorized deckers.
 - The CDZ boundary will be clearly marked.
 - Safety deckers in a CDZ are not required to be tied off until they are no longer engaged in leading-edge work.
- **Rescue Plan:** In the event of a fall, a prompt rescue plan is in place. This includes having a lift (e.g., aerial lift) on standby, trained personnel, and emergency contact information readily available. Suspension trauma will be addressed in training.

4. Erection Stability & Bracing Plan

The structure must be stabilized at all times during erection.

- **Anchor Bolts:** All columns must be erected on a foundation with a minimum of four (4) anchor bolts. No column shall be released from the crane until it is secured with at least four bolts.
- **Initial Connections:** Before releasing the crane load, beams and girders must be secured with a minimum of two (2) bolts per connection, installed in accordance with OSHA 1926.756(c)(1).
- **Temporary Bracing:** The structure will be stabilized with temporary guys, cables, or structural members as detailed in the erection drawings. No more than eight stories of steel shall be erected before the structure is fully plumbed and bolted. For this 3-storey structure, the frame must be plumbed and stabilized before proceeding to the next tier.
- **Diagonal Bracing:** Permanent diagonal bracing will be installed as soon as practical. These members are critical for structural integrity.
- **Load Placement:** No construction loads (e.g., bundles of decking, equipment) are allowed on the unbraced steel frame. Decking may only be placed on a fully bolted and braced bay.

5. Hazard Register & Control Measures

Hazard	Potential Harm	Control Measures
Crane Collapse / Dropped Load	Fatalities, serious injuries, property damage	Use correct crane for the load/radius, verify load charts, ensure stable ground conditions, conduct daily inspections, use a qualified operator/rigger.
Falls from Height	Fatalities, serious injuries	100% fall protection (PFAS) over 30 ft (15 ft for non-connectors), proper anchor points, guardrails, CDZ procedures, training.
Struck-by/Caught-between	Fatalities, crush injuries, amputations	Controlled access zones (crane swing area), use of taglines, clear communication, no work under suspended loads, situational awareness.

Hazard	Potential Harm	Control Measures
Structural Collapse	Multiple fatalities, catastrophic failure	Follow erection drawings and sequence, install temporary bracing, meet minimum bolt requirements before unhooking, no overloading of frame.
Adverse Weather	Dropped loads (wind), electrocution (lightning)	Monitor weather continuously, adhere to wind speed limits, suspend operations during lightning or high winds.
Improper Rigging	Dropped load, rigging failure	Use only a Qualified Rigger, inspect all rigging daily, use appropriate rigging for the load (spreader bar), protect slings from sharp edges.

6. PPE & Site Access Notes

6.1. Personal Protective Equipment (PPE):

- Standard PPE (100% of the time):
- Hard Hat (ANSI Z89.1)
- Safety Glasses with side shields (ANSI Z87.1)
- High-Visibility Vest or Shirt
- Steel-Toed Work Boots
- Task-Specific PPE:
- Fall Protection: Full-body harness and appropriate connecting device when working at height.
- Gloves: Heavy-duty work gloves for material handling and connecting.
- Hearing Protection: In areas where noise levels exceed OSHA action levels.

6.2. Site Access Control:

- Controlled Access Zone (CAZ): A CAZ will be established around the crane's operational footprint and directly under all overhead erection work.
- Barricades & Signage: The CAZ will be clearly marked with "DANGER - OVERHEAD WORK" signs and barrier tape or fencing.

- Authorized Personnel: Only trained and authorized personnel directly involved in the steel erection process are permitted inside the CAZ. All others must obtain permission from the site supervisor before entering.

Plan Acknowledgment & Approval

This plan has been reviewed and is approved for implementation. All personnel involved in steel erection activities will be trained on the contents of this plan.

Qualified Person: _____ (Print Name)

Signature: _____

Date: _____